

# TARZAN PACKCELL

## Embodiment RFQ & Application Engineering Packet v1

Teach once. Execute on changed orders.

|                |                                                                                                                    |
|----------------|--------------------------------------------------------------------------------------------------------------------|
| REQUEST        | Written validation of a commercially defensible single-arm, jaw/suction packing embodiment.                        |
| FROZEN GATES   | 106 mm minimum usable jaw opening; 8 kg workpiece; 20% arm payload margin; 3 mm insertion clearance.               |
| CURRENT RESULT | No eligible arm-tool pair is frozen. UR15 is the conditional reach/payload preference; tooling remains unresolved. |

### Purpose

Tarzan is selecting a single-arm embodiment for a teach-once small-sortable ecommerce packing cell. This packet requests manufacturer and integrator application engineering. It is not a purchase order and does not relax the frozen task envelope.

12 July 2026 | Tarzan Engineering

## 1. Frozen task envelope

| Requirement                 | Requirement                                           |
|-----------------------------|-------------------------------------------------------|
| Development layouts         | 50 deterministic layouts; confirmation layouts sealed |
| Maximum reach proxy         | 1.148 m including current tool-length assumption      |
| Carton interior             | 400 x 300 x 250 mm                                    |
| Representative objects      | 50 x 50 x 50; 120 x 80 x 60; 250 x 150 x 100 mm       |
| Maximum workpiece           | 8.0 kg                                                |
| Jaw opening                 | 106 mm minimum; 110-120 mm preferred                  |
| Payload margin              | 20% on tool plus workpiece                            |
| Workpiece CoG offset        | Up to 200 mm in current selection model               |
| Current static moment proxy | 23.1 Nm before custom-finger mass and acceleration    |
| Insertion clearance         | 3 mm minimum                                          |
| Object force limit          | 80 N unless metadata explicitly permits more          |

### Provisional performance target

Quote against 1.5 m/s<sup>2</sup> nominal acceleration, 2.0 m/s<sup>2</sup> peak, and 12 seconds per item. These are commercial targets for application review, not validated robot limits. Return a manufacturer-supported acceleration and cycle estimate.

### Current reach proxy

| Arm  | Max required | Min margin | Derated mass | Status                 |
|------|--------------|------------|--------------|------------------------|
| UR15 | 1.148 m      | 152 mm     | 11.4 kg      | Conditional preference |
| UR20 | 1.148 m      | 602 mm     | 11.4 kg      | Use only if required   |

## 2. Arm application questions

- Confirm payload and CoG validity for tool plus 8 kg workpiece with 20% margin.
- Confirm wrist force/torque validity at maximum extension and declared CoG.
- Provide acceleration, speed, orientation, and stopping-distance derating.
- Identify unreachable, singular, collision-limited, or reduced-performance poses.
- Estimate cycle time and state whether UR20 materially improves the application.
- Describe force/torque, safety, service, and controls integration.
- Provide CAD, kinematic, inertial, collision, payload-curve, and interface data.

## 3. End-effector application questions

### Custom-finger OnRobot 2FG14

The standard 105 mm external opening does not pass. Propose a manufacturer-supported configuration providing at least 106 mm, preferably 110-120 mm, and validate:

- Minimum closure for 30-50 mm objects.
- Finger dimensions, mass, CoG, inertia, bending moment, and permitted materials.
- Force transmission and allowable acceleration with an 8 kg workpiece.
- Compliance with documented 25 Nm Y and 30 Nm X platform limits.
- Controllable force range for fragile packaged goods.
- Carton-entry and box-rim collision envelope.
- CAD, inertial, collision, controls, revision, and licensing data.

### Hybrid jaws and suction

Identify a supported vacuum or hybrid architecture for flat sealed cardboard/plastic parcels up to 8 kg. Provide cup geometry, required seal area, porosity limits, vacuum reserve, acceleration derating, tool mass/CoG, seal-loss diagnostics, tool-change time, and safe recovery. Unknown surfaces must fall back to jaws or human handling.

## 4. Required simulation and integration data

- Official CAD with revision identifiers and licensing terms.
- URDF or equivalent kinematic description where available.
- Link/tool masses, centers of gravity, and inertias.
- Collision geometry distinct from visual geometry.
- Joint, actuator, force, speed, acceleration, and payload limits.
- TCP/contact frames and finger or suction contact geometry.
- Control protocol documentation and integration examples.
- Permission for internal simulation, derived MJCF generation, and benchmark publication.

## 5. Requested written outcome

Return one of the following, with every boundary condition stated:

- UR15 valid across the declared envelope.
- UR20 required, with the precise disqualifying UR15 condition.
- Custom 2FG14 valid, with a reviewed finger design.
- Hybrid jaw/suction architecture required.
- No supported configuration.

**Tarzan will not relax the 106 mm or 8 kg gates to fit a preferred component. Unsupported items must be escalated rather than attempted.**

### Attachments available

- Full 50-layout waypoint artifact.
- Coupled embodiment decision matrix.
- Hybrid portfolio specification.
- Light-sortable MVP specification.
- Panda negative embodiment evidence and anti-cheat boundary.